METRIC

MIL-PRF-29595/1(AS)
w/AMENDMENT 1
05 May 2011
SUPERSEDING
MIL-PRF-29595/1(AS)
26 April 2011

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 26-VOLT, 25-AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-29595.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be M29595/1-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 contained herein. The weight of the battery shall be not greater than 11.8 kilograms.

3. Capacity. The initial capacity of the battery shall be 32 ampere-hours (Ah). The rated capacity of the battery shall be 25 ampere-hours (Ah) at the end of life. Each battery shall produce a potential of not less than 20.0 volts at not less than 60 minutes when connected to a 25.0-ampere load after being tested in accordance with 4.5.8 as modified herein at 24° C (75° F) (25 Ah/1.0 hr/24° C/20.0V at end of life). See 3.9.9 as modified herein.

4. Aircraft installation. For installation in the aircraft, the battery shall be supported as shown on figure 3 of MIL-PRF-29595. Vibration isolation shall be provided to ensure that the vibration level does not exceed the levels of the 2-year accelerated life test of 4.5.11 as modified herein. If vibration isolators are used, sufficient sway clearance between the battery and the mounting structure shall be provided.

5. Qualification, conformance and safety tests inspection. Table I of this specification sheet lists the required qualification and conformance inspections. Table II lists the required S9310-AQ-SAF-010 tests to be performed before the manufacturer can receive safety certification approval and qualified products database (QPD) listing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

6. MIL-PRF-29595 variance. The battery shall comply with MIL-PRF-29595 except as follows.

6.1 The following paragraphs are not applicable:

3.7.2 Polarity marking.
3.9.4 Dielectric strength and insulation resistance.
3.9.7 Built-in test.
3.9.11 Strength of receptacle.
3.9.24 Cold and hot temperature charge.
3.9.26 Transient drop test.
4.5.5 Dielectric strength and insulation resistance test.
4.5.10 Strength of receptacle test.
4.5.13 Charge and discharge test at low temperature.
4.5.23 Cold and hot temperature charge test.
4.5.24 Wormgear retaining pin hardness.
4.5.27 Transient drop test.
4.5.34 Fluid susceptibility.

6.2 Modify the following paragraph of MIL-PRF-29595:

2.2.1 Specifications and standards. Add the following documents:

DEPARTMENT OF DEFENSE SPECIFICATIONS:

"MIL-DTL-38999	-	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
" MS27466	-	Connectors, Receptacle, Electrical, Wall Mounting Flange, Crimp Type, Bayonet Coupling, Series I"

3.5 Design. Delete the text and insert: "The design shall conform to the requirements specified by MIL-PRF-29595, and MIL-PRF-29595/1 when examined in accordance with the visual and mechanical tests of 4.5.2 of MIL-PRF-29595. The design shall also conform to the requirements specified in S9310-SAF-AQ-010 (see 6.17 and 6.18 both of MIL-PRF-29595).

3.5.3 Battery container and cover. Delete the text in the second sentence and insert: "The dimensions and locations of the receptacle, vent tubes, and battery identification shall be as shown on figure 1 herein.

3.5.5 Receptacles. Insert the following after the first sentence: "The battery's electrical interface to the aircraft shall be in accordance with MIL-DTL-38999, series III and equivalent to

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

the MS27466T17B6S receptacle. It shall be positioned and electrically connected as shown on figures 1 and 2, respectively herein.

"a. Pin A shall be high side AC (input heater power).

"b. Pin B shall be AC ground (heater return).

"c. Pin C shall be battery negative.

"d. Pin D shall be battery positive.

3.5.9 Heaters. In the first sentence, delete "may" and insert "shall". In 3.5.9.a, delete the text and insert the following: "Heater shall be powered by 115 volts alternating current through pin A and pin B of the MS27466T17B6S receptacle." In 3.5.9.b, delete the text and insert the following: "Heater current shall be not greater than 3.0 amperes. The rated current of each heater control device shall be compatible with the heater current."

3.7.1 Color. Delete 3.7.1, 3.7.1.a, 3.7.1.b, and 3.7.1.c and insert: "The color under natural light of all visible parts of each battery, except the mounting points, other external hardware, identification marking, and instructions, shall conform to color 17875 (white), in accordance with FED-STD-595.

3.7.4 Battery caution marking. Delete the text in the first sentence and insert the following: "The battery caution marking shall contain the required titles and information as shown in figure 2 of MIL-PRF-29595 and shall be placed as marking #2 shown on figure 1 herein."

3.7.5 Marking for batteries with heaters. In the first sentence, delete "battery identification marking and battery caution marking" and insert "battery cover and the surface bearing the battery identification". After the third sentence, insert: "(see figure 1 of this specification sheet)."

3.9.6 State of charge indicator. Delete the text and insert: "A state of charge device with indicator shall be as shown on figure 1 herein."

3.9.8 Open circuit voltage. Delete the text and insert: "The open circuit voltage shall not be greater than 29 volts when tested in accordance with 4.5.4 of MIL-PRF-29595."

3.9.9 Capacity and electrical performance. Delete the text of paragraph in its entirety and insert: "The initial capacity of the battery shall be not less than 32.0 ampere-hours. The rated capacity (see 6.14.1 of MIL-PRF-29595) of the battery shall be not less than 25 ampere-hours. To meet the capacity requirement, each battery shall produce a potential of not less than 20.0 volts at not less than 60 minutes when connected to a 25 ampere load after being tested in accordance with 4.5.8 as modified herein at 21° C (70° F) (25 Ah/1.0 hr/21°C/20.0 V at end of

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

life). The battery's 1C-rate shall be 25 amperes."

3.9.12 Cycling. Delete the text of paragraph in its entirety and insert: "Batteries shall successfully complete not less than 600 cycles for qualification inspection and 100 cycles for conformance inspection after being tested in accordance with 4.5.11 as modified herein. The battery shall deliver not less than 25 ampere-hours when discharged at the 1.0 C-rate for 1 hour. The inability of the battery to deliver 25 ampere-hours for three consecutive discharges shall constitute a failure."

3.9.20 Deep discharge recovery. Delete the text of the paragraph in its entirety and insert: "The battery may use protective circuitry to protect it from deep discharge. If the battery uses protective circuitry, the battery shall meet the requirements of 3.9.20.a. Otherwise, the battery shall meet the requirements of 3.9.20.b.

"a. If utilizing protective circuitry (the battery while under load), the protection circuitry shall allow the battery to discharge down to 16.0 volts before stopping the discharge. If protective circuitry is utilized to stop the discharge, there must be a time delay of 30 seconds before the battery is removed from the DC bus. If the battery goes into open circuit at 16.0 volts, the battery shall stay off line (no on-off-on-off chatter) until a charge voltage of 24.0 volts or greater is applied to the battery terminals.

"b. If protective circuitry is not utilized, the battery shall be capable of storage in the fully discharged state (0.0 volts) (see 6.14.5 of MIL-PRF-29595) in uncontrolled temperature conditions from -40° C to +50° C (-40° F to +122° F) for not less than 168 ±2 hours without damage. After such storage, the battery shall deliver not less than 25 ampere-hours after being tested in accordance with 4.5.20 as modified herein. The battery shall also meet the requirements of 3.8.1 of MIL-PRF-29595 during and after charge and discharge."

3.9.22 Output performance. Delete the text and insert the following: "The battery shall produce a potential of not less than 20.0 volts during and after the test of 4.5.21 as modified herein and the battery shall produce not less than 25 ampere-hours when discharged in accordance with 4.5.21.c of MIL-PRF-29595."

3.9.23 Discharge and charge at 71° C . Delete the text in the first sentence and insert the following: "The battery shall produce not less than 25 ampere-hours when discharged in accordance with 4.5.22.c. of MIL-PRF-29595."

3.9.25 Vibration. Delete the text in the first sentence and insert the following: "During the random vibration tests of 4.5.26 as modified herein, the fluctuation of the battery voltage shall be not greater than 1.5 volts of the load voltage and shall not fall below the minimum closed circuit voltage of 20.0 volts."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

6.3 Add the following paragraphs:

"3.9.28 Aircraft start-up load. The battery shall produce a potential of not less than 20.0 volts during and after the tests of 4.5.40.1 as specified herein."

"3.9.29 Aircraft emergency load. The battery shall produce a potential of not less than 20.0 volts during and after the tests of 4.5.40.2 as specified herein."

"3.9.30 Half-hour charging. The battery shall produce not less than 20 ampere-hours after being tested in accordance with 4.5.40.3.c as specified herein. The battery shall produce not less than 20 ampere-hours after being tested in accordance with 4.5.40.3.h as specified herein. The battery shall also meet the requirements of 3.8.1 of MIL-PRF-29595."

"3.9.31 One-hour charging. The battery shall produce not less than 20 ampere-hours after being tested in accordance with 4.5.40.4.e as specified herein. The battery shall not sustain any damage when a 28.25 ±0.25 volt charging source is applied continuously to the battery during the test of 4.5.40.4 as specified herein. The battery shall also meet the requirements of 3.8.1 of MIL-PRF-29595."

6.4 Modify the following paragraphs:

4.2 Qualification inspection. Delete the text in the first sentence and insert the following: "Qualification inspection shall consist of the examinations and tests specified in table I of this specification sheet (see 6.4 as modified herein and 6.5 of MIL-PRF-29595)."

4.3.5 Group B inspection. Delete the text and insert the following: "The group B test shall consist of the examinations and tests specified in table I of this specification sheet (see 6.8 of MIL-PRF-29595)."

4.5 Methods of inspection. Delete the text in the first sentence and insert the following: "The tests on each battery shall be performed in the order as specified in table I of this specification sheet."

4.5.8 Capacity Discharge. Delete the text of the paragraph in its entirety and insert: "Stabilize the battery at room temperature (see 4.4.1 of MIL-PRF-29595). Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts. Afterwards, constant potential charge the battery at 28.25 ±0.25 volts for 2 hours. Examine the battery for the requirements of 3.9.9 as modified herein and requirement (1) of table III of this specification sheet."

4.5.11 Cycling test. Delete the text and insert: “For qualification inspection, subject the battery to 600 cycles of discharge-charge-open circuit and to the temperature environments in accordance with 4.5.11.1 as specified herein. For conformance inspection, subject the battery to 100 cycles of discharge-charge-open circuit in accordance with 4.5.11.2 as specified herein.”

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

6.5 Add the following paragraphs:

"4.5.11.1 Qualification inspection. Simultaneously subject the battery to 600 cycles of discharge-charge-open circuit (rest) and to the temperature environments specified in the following paragraphs. This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2 of MIL-PRF-29595.

"b. First sequence (cycles 1 through 100).

"(1) Cold temperature environment:

"(a) Stabilize the battery temperature at -18° C (0° F). Perform steps 4.5.11.1.b(1)(b) through 4.5.11.1.b(1)(e) with the battery in the -18° C (0° F) temperature environment.

"(b) Discharge the battery at the 1.0 C-rate for 1 hour.

"(c) Immediately after completing step 4.5.11.1.b(1)(b), constant potential charge the battery at 28.25 ±0.25 volts for 2 hours.

"(d) Place the battery on open circuit for a period of 2.0 ±0.5 hours.

"(e) Repeat above steps 4.5.11.1.b(1)(b) through 4.5.11.1.b(1)(d) for a total of 100 cycles.

"c. Second sequence (cycles 101 through 150).

"(1) Hot temperature environment:

"(a) Stabilize the battery temperature at 43° C (110° F). Perform steps 4.5.11.1.c(1)(b) through 4.5.11.1.c(1)(e) with the battery in the 43° C (110° F) temperature environment.

"(b) Discharge the battery at the 1.0 C-rate for 1 hour.

"(c) Immediately after completing step 4.5.11.1.c(1)(b), constant potential charge the battery at 28.25 ±0.25 volts for 2 hours.

"(d) Place the battery on open circuit for a period of 2.0 ±0.5 hours.

"(e) Repeat above steps 4.5.11.1.c(1)(b) through 4.5.11.1.c(1)(d) for a total of 50 cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

"d. Third sequence (cycles 151 through 250).

"(1) Ambient temperature environment:

"(a) Stabilize the battery temperature at 24° C (75° F). Perform steps 4.5.11.1.d(1)(b) through 4.5.11.1.d(1)(e) with the battery in the 24° C (75° F) temperature environment.

"(b) Discharge the battery at the 1.0 C-rate for 1 hour.

"(c) Immediately after completing step 4.5.11.1.d(1)(b), constant potential charge the battery at 28.25 ± 0.25 volts for 2 hours.

"(d) Place the battery on open circuit for a period of 2.0 ±0.5 hours.

"(e) Repeat above steps 4.5.11.1.d(1)(b) through 4.5.11.1.d(1)(d) for a total of 100 cycles.

"e. Fourth sequence (cycles 251 through 300).

"(1) Hot temperature environment:

"(a) Stabilize the battery temperature at 65° C (149° F). Perform steps 4.5.11.1.e(1)(b) through 4.5.11.1.e(1)(e) with the battery in the 65° C (149° F) temperature environment.

"(b) Discharge the battery at the 1.0 C-rate for 1 hour.

"(c) Immediately after completing step 4.5.11.1.e(1)(b), constant potential charge the battery at 28.25 ±0.25 volts for 2 hours.

"(d) Place the battery on open circuit for a period of 2.0 ±0.5 hours.

"(e) Repeat above steps 4.5.11.1.e(1)(b) through 4.5.11.1.e(1)(d) for a total of 50 cycles.

"f. Repeat steps 4.5.11.1.b through 4.5.11.1.e for a total of 600 cycles.

"g. Stabilize the battery temperature at 24° C (75° F) and discharge at the 1C-rate to 20.0 volts.

"h. Examine the battery for the requirements of 3.9.12 as modified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

"4.5.11.2 Conformance inspection. Constant potential charge the battery in accordance with 4.4.2 of MIL-PRF-29595 at room temperature. Subject the battery to 100 cycles of discharge-charge-open circuit (rest) at ambient condition of 4.4.1 of MIL-PRF-29595. Each cycle shall consist of 1.0-hour, 1.0 C-rate discharge, followed immediately by a 2.0-hour constant potential charge in accordance with 4.4.2, followed by a 1.0-hour rest period. Examine the battery for the requirements of 3.9.12 as modified herein."

6.6 Modify the following paragraphs:

4.5.14 Discharge while inverted. Delete 4.5.14.c in its entirety and insert the following: "c. Discharge the battery at the 2.5C-rate for 5 minutes or to a cutoff voltage of 20.0 volts, whichever occurs first. At 2.5 minutes into the discharge, place the battery upright."

4.5.15 Temperature rise and float test. Delete the text of 4.5.15 and all of its subparagraphs and insert: "This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2.

"b. Place the battery in a temperature chamber at 49° C (120° F) for not less than 12 hours. Perform the following with the battery in the 49° C (120° F) temperature environment.

"c. Discharge the battery at the 2.5C-rate for 5 minutes or to a cutoff voltage of 20.0 volts, whichever occurs first.

"d. Immediately after the 2.5C-rate discharge and with the battery still in the temperature chamber, constant potential charge the battery at 28.6 ±0.10 volts for 24 ±0.1 hours.

"e. Stabilize the battery temperature at room temperature (see 4.4.1 of MIL-PRF-29595). Discharge the battery at the 1.0 C-rate to a cut-off voltage of 20.0 volts.

"f. Measure voltage, current, and time throughout the test.

"g. Remove the battery from the temperature chamber.

"h. Examine the battery for the requirements of 3.9.15 of MIL-PRF-29595 and requirement (1) of table III of this specification sheet."

4.5.18 Physical integrity at high temperature test. In 4.5.18.g, delete the text "4.0C" and insert "2.5C."

4.5.20 Deep discharge recovery test. Delete 4.5.20.d and insert "Constant potential charge the battery at 28.25 ±0.25 volts for 10 hours."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

4.5.21 Output performance. Delete 4.5.21.g and insert the following: "g. Discharge the battery in accordance with figure 6 herein to a cutoff voltage of 20.0 volts. Examine the battery for the requirement (3) of table III of this specification sheet."

4.5.22 Discharge and charge at 71°C. Delete 4.5.22.c and insert the following: "c. With the battery still in the temperature chamber at 71°C (160 °F), discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts."

4.5.26 Vibration test. Delete the text of 4.5.26.d in its entirety and insert the following: "d. Energize the battery heaters during vibration. Vibrate the battery along each orthogonal axis (in the axis order shown on figure 3 of MIL-PRF-29595) in accordance with the test levels on figure 3 of this specification sheet. Subject the battery to two 15-minute logarithmic sweeps from 5-500-5 Hertz. Conduct one 5-minute logarithmic sweep from 20-30-20 hertz. Conduct a resonance search from 5-500 hertz. Perform resonance dwell testing for 1.5 minutes at each resonance, to a maximum of three most severe exhibiting a transmissibility of two or greater. Vibrate the battery for a period of 20 minutes per axis in accordance with level 1 on figure 4 of this specification sheet. Vibrate the battery for a period of 5 minutes per axis to level 2 on figure 4 of this specification sheet." In 4.5.26.g delete the text "4.0C" and insert "2.5C".

4.5.28 Shock test (basic design). In 4.5.28.f delete the text "4.0C" and insert "2.5C".

4.5.29 Shock test (crash safety). In 4.5.29.f delete the text "4.0C" and insert "2.5C".

4.5.30 Altitude. Delete the text of 4.5.30 and all of its subparagraphs and insert: "This test consists of the following steps.

"a. Constant potential charge the battery at 28.25 ±0.25 volts for 2 hours.

"b. Immediately after completion of charge, place the battery inside an environmental chamber.

"c. Within 15 minutes after completing step 4.5.30.b, lower the chamber temperature to -29° C (-20° F) and lower the pressure inside the chamber to simulate the ambient pressure at 18,288 meters (60,000 feet) altitude. Perform the remainder of the test with the battery in the temperature environment.

"d. Discharge the battery at the 2.5C-rate for 3 minutes. Examine the battery for the requirements of 3.9.13 of MIL-PRF-29595.

"e. Constant potential charge the battery at 28.25 ±0.25 volts for 2 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

"f. Open-circuit the battery and, within 5 minutes, return the chamber to ground ambient air pressure (see 4.4.1). Discharge the battery at the 2.5C-rate for 3 minutes. Examine the battery for the requirements of 3.9.13 of MIL-PRF-29595.

"g. Repeat step 4.5.30.c. Repeat steps 4.5.30.e, and 4.5.30.f.

"h. Examine the battery for the requirement of 3.8.1 of MIL-PRF-29595".

4.5.31 Humidity test. In 4.5.31.k delete the text "4.0C" and insert "2.5C".

4.5.32 Salt fog. In 4.5.32.j delete the text "4.0C" and insert "2.5C".

4.5.33 Temperature shock test. In 4.5.33.j delete the text "4.0C" and insert "2.5C".

4.5.37 Shelf life test. Delete the text of 4.5.37 in its entirety and insert the following: "Constant potential charge the battery in accordance with 4.4.2 of MIL-PRF-29595. Store the battery for not less than 18 months in an uncontrolled temperature environment (such as an unheated, uncooled warehouse) where the ambient air temperature shall be within the range of -40° C to 38° C (-40° F to 100° F). Bring the battery to the test facility. Constant potential charge the battery at 28.25 ± 0.25 volts for 10 hours. Stabilize the battery at room temperature (see 4.4.1 of MIL-PRF-29595). Discharge the battery at the C-rate to 20.0 volts. Constant potential charge the battery at 28.25 ±0.25 volts for 10 hours. Subject the battery to the cycling test of 4.5.11 as modified herein. Examine the battery for the requirements of 3.9.12, as modified herein and 3.9.19 of MIL-PRF-29595."

4.5.38 Penetration test. Delete the text in the first sentence and insert the following: "The battery shall be tested in accordance with step (a)."

4.5.40 Special tests. Delete the text and insert the following: "Each of the three battery samples shall be subjected to each of the following tests."

6.7 Add the following paragraphs:

"4.5.40.1 Aircraft start-up loads. This test consists of the following steps.

"a. Stabilize sample 1 at 55° C (131° F), stabilize sample 2 at -29° C (-20° F), and stabilize sample 3 at 25° C (77° F) for 12 hours.

"b. With each battery still in the temperature chamber at the indicated test temperature, discharge each sample in accordance with figure 5 of this specification sheet.

"c. Examine each sample for the requirements of 3.9.28 of this specification sheet."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

"4.5.40.2 Aircraft emergency loads. This test consists of the following steps.

"a. Discharge each sample in accordance with step 1 of figure 6 of this specification sheet.

"b. Constant potential charge each sample in accordance with 4.4.2 of MIL-PRF-29595.

"c. Stabilize each sample battery's temperature at the test temperature specified below. With each sample still in the temperature chamber at the indicated test temperature, discharge each sample as follows:

"(1) Sample 1 in accordance with step 2 of figure 6 of this specification sheet.

"(2) Sample 2 in accordance with step 3 of figure 6 of this specification sheet.

"(3) Sample 3 in accordance with step 4 of figure 6 of this specification sheet.

"d. Constant potential charge each sample in accordance with 4.4.2 at room temperature.

"e. Discharge sample 1 in accordance with step 5 of figure 6 of this specification sheet.

"f. Constant potential charge sample 1 in accordance with 4.4.2 at room temperature.

"g. Examine each sample for the requirements of 3.9.29 of this specification sheet."

"4.5.40.3 Half-hour charge test. This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2.

"b. Stabilize sample 1 at -18°C (0°F), stabilize sample 2 at 15° C (59° F), and stabilize sample 3 at 55° C (131° F). Perform 4.5.40.3.c through 4.5.40.3.g of this specification sheet with the battery in the environmental chamber at the specified temperature.

"c. Discharge each sample at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"d. Examine the battery for the requirements of 3.9.30 of this specification sheet.

"e. Stabilize the samples at 25° C (77° F). Constant potential charge the battery in accordance with 4.4.2.

"f. Stabilize the samples at the temperatures of 4.5.40.3.b. Discharge each sample at the 1.0 C-rate to remove all but 14 ampere-hours from each sample as determined by 4.5.40.3.c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

"g. Stabilize the samples at the temperatures of 4.5.40.3.b. Constant potential charge the battery at 28.25 ±0.25 volts for 30 minutes.

"h. Immediately after completing 4.5.40.3.g, place the samples in a room temperature environment (see 4.4.1 of MIL-PRF-29595). Discharge the samples at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"i. Examine the battery for the requirements of 3.9.30, as specified herein."

"4.5.40.4 One-hour charge test. This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2.

"b. Stabilize the battery at -40° C (-40° F). Perform 4.5.40.4.c and 4.5.40.4.d with the battery in the environmental chamber at the specified temperature.

"c. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"d. Stabilize the battery (see 4.4.1 of MIL-PRF-29595) at -40° C (-40° F). Constant potential charge the battery in accordance with 4.4.2 for 1.0 hour.

"e. Immediately after completing 4.5.40.4.d, place the battery in a room temperature environment (see 4.4.1) and discharge at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"f. Examine the battery for the requirements of 3.9.31 as specified herein."

4.5.41 Final examination. Delete the second sentence.

6.8 Modify the following paragraph:

6.4 Qualification. Insert the following after the first sentence: "Batteries must have successfully completed the safety test requirements of S9310-AQ-SAF-010 and received safety certification approval before being listed on the QPD."

6.9 Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

TABLE I. Qualification inspection and conformance inspection of batteries. 1/

Test number	Examination and tests	Qualification inspection							Conformation inspection		
		Sample number					Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3	4	5				1	2
1	Incoming inspection	X	X	X	X	X	3.9.1 (G)	4.5.1 (G)		X	X
2	Visual and mechanical examination	X	X	X	X	X	3.9.2 (G)	4.5.2 (G)	X	X	X
3	Dimensions and weight	X	X	X	X	X	3.9.3 (G)	4.5.3 (G)	2/	X	
4	Strength of vent tubes test	X	X	X	X	X	3.9.5 (G)	4.5.6 (G)	2/	X	X
5	Electrical test preparation	X	X	X	X	X	---	4.5.7 (G)		X	X
6	Capacity discharge	X	X	X	X	X	3.9.9 (S), reqmt 3 (N)	4.5.8 (S)	X	X	X
7	Aircraft start-up loads	X	X	X			3.9.28 (N)	4.5.40.1 (N)			
8	Aircraft emergency loads	X	X	X			3.9.29 (N)	4.5.40.2 (N)			
9	Half-hour charge	X	X	X			3.9.30 (N)	4.5.40.3 (N)			
10	One-hour charge			X			3.9.31 (N)	4.5.40.4 (N)			
11	Cycling test	X					3.9.12 (S)	4.5.11 (S)		X	
12	Output performance		X				table III reqmt (3) (G)	4.5.21 (S)			X
13	Discharge while inverted		X				3.6.2 (G), table III reqmt (4) (G)	4.5.14 (S)			X
14	Temperature rise and float		X				3.9.15 (G)	4.5.15 (S)			X
15	Thermal protection			X			3.8.2 (G)	4.5.17 (G)			
16	Physical integrity at high temperature				X		3.8.1 (G), table III reqmt (1) (G)	4.5.18 (S)			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

TABLE I. Qualification inspection and conformance inspection of batteries - Continued. 1/

Test number	Examination and tests	Qualification Inspection							Conformation inspection		
		Sample number					Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3	4	5				1	2
17	Impact resistance				3/		3.9.21 (G)	4.5.19 (G)			
18	Deep discharge recovery			X			3.9.20 (S)	4.5.20 (S)			X
19	Discharge and charge at 71 °C (160 °F)				X		3.9.23 (S)	4.5.22 (S)			
20	Shock test (basic design)			X			3.8.1 (G), 3.9.27 (G)	4.5.28 (S)		X	
21	Altitude test		X				3.6.1.2 (G), 3.8.1 (G)	4.5.30 (S)			X
22	Low pressure (altitude)		X				3.6.1.2 (G), 3.8.1 (G), 3.9.3 (G)	4.5.25 (G)			
23	Humidity test		X				3.6.1.1 (G), 3.8.1 (G), 3.9.13 (G)	4.5.31 (S)			X
24	Salt fog test	X					3.8.1 (G), 3.9.13 (G)	4.5.32 (S)		X	
25	Temperature shock	X					3.8.1 (G), 3.9.13 (G)	4.5.33 (S)		X	
26	Sand and dust					X	3.8.1 (G), 3.9.8 (S), 3.9.9 (S)	4.5.35 (G)			
27	Vibration			X			3.8.1 (G), 3.9.3 (G), 3.9.25 (S)	4.5.26 (S)		X	
28	Shock test (crash safety)			X			3.9.13 (G), 3.9.27 (G)	4.5.29 (G)			
29	Crush test			X			3.9.17 (G)	4.5.39 (G)			
30	Electromagnetic interference				X		3.9.14 (G)	4.5.12 (G)			
31	Penetration				X		3.9.16 (G)	4.5.38 (S)			
32	Ground storage					X	3.9.18 (G)	4.5.36 (G)			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

TABLE I. Qualification inspection and conformance inspection of batteries - Continued. 1/

Test number	Examination and tests	Qualification inspection							Conformation inspection		
		Sample number					Requirement paragraph	Test Paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3	4	5				1	2
33	Shelf life					X	3.9.12 (S), 3.9.19 (G)	4.5.37 (S)			
34	Final examination	X	X	X	X	X	3.9.4 (G)	4.5.41 (S)		X	X

1/ (G) indicates that paragraph is in MIL-PRF-29595. (S) Indicates that paragraph has been modified by specification sheet. (N) Indicates new requirement added by specification sheet.
2/ See sample information in test paragraph.
3/ For nonmetallic battery container designs, two out nonmetallic battery containers shall be furnished for inspection.

TABLE II. Safety test inspection (S9310-AQ-SAF-010 requirements) list. 1/

Test number	Examination and tests	Sample number	Requirement paragraph	Test paragraph
1	Penetration test	1 each minimum	3.9.16 (G)	4.5.38 (S)
2	Crush test	1 each minimum	3.9.17 (G)	4.5.39 (G)
3	Short Circuit Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
4	Overcharge/Discharge Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
5	Overdischarge/Charge Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
6	High Temperature Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
7	Electrical Safety Device Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010
8	Aging Safety Test	1 each minimum	See S9310-AQ-SAF-010	See S9310-AQ-SAF-010

1/ (G) indicates that paragraph is in MIL-PRF-29595. (S) Indicates that paragraph has been modified by a specification sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

TABLE III. Capacity performance requirements.

Requirement Number	Discharge Cutoff Volts & Temperature (see 4.4.1)	Rate of Discharge Factor	Minimum Time to Cutoff Voltage
(1)	20.0 volts at room temperature	1.0C	1 hour
(2)	20.0 volts at 49° C	1.0C	1.1 hours
(3)	20.0 volts at –29° C	1.0C	45 minutes
(4)	20.0 volts at room temperature	2.5C	5 minutes

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

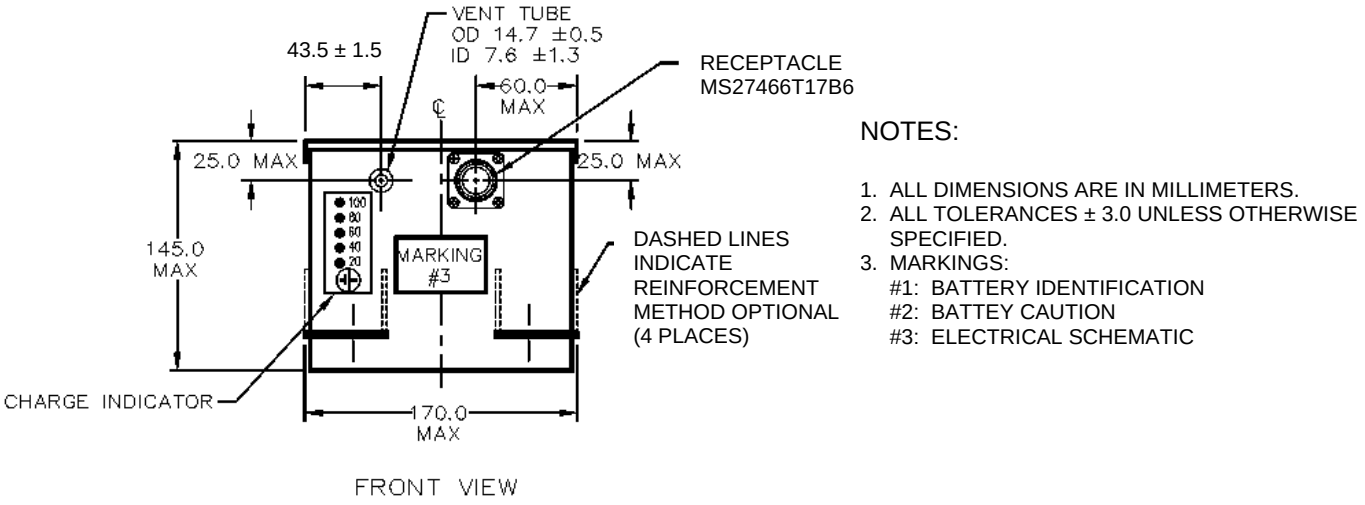
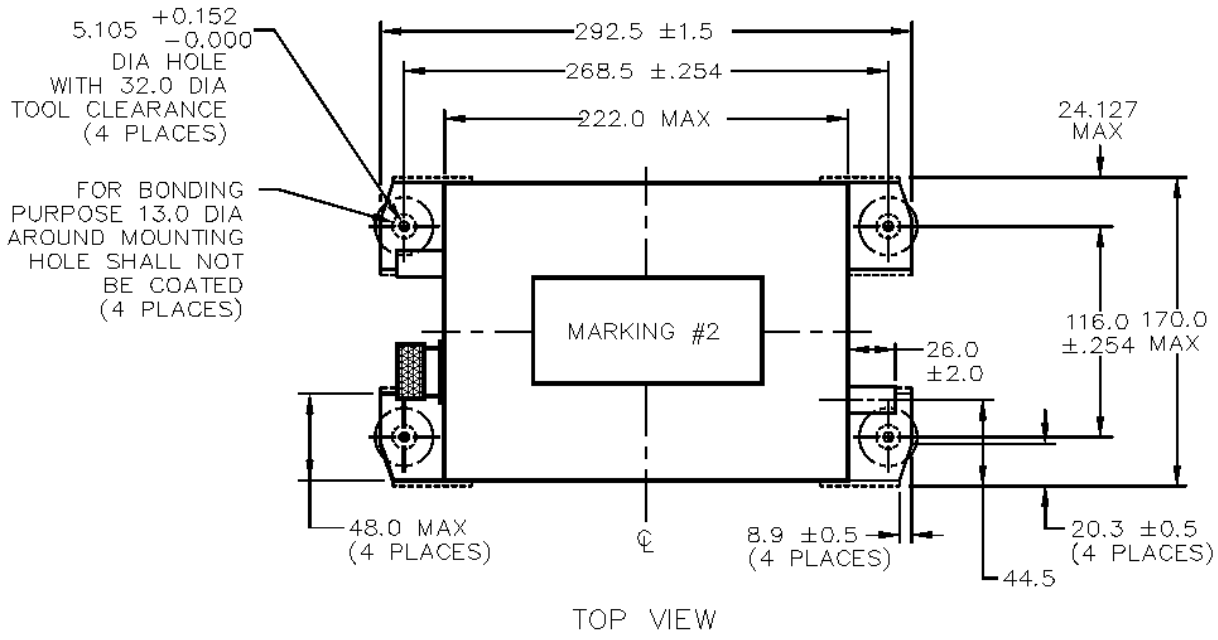


FIGURE 1. Dimensions and configuration for PIN M29595/1-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.



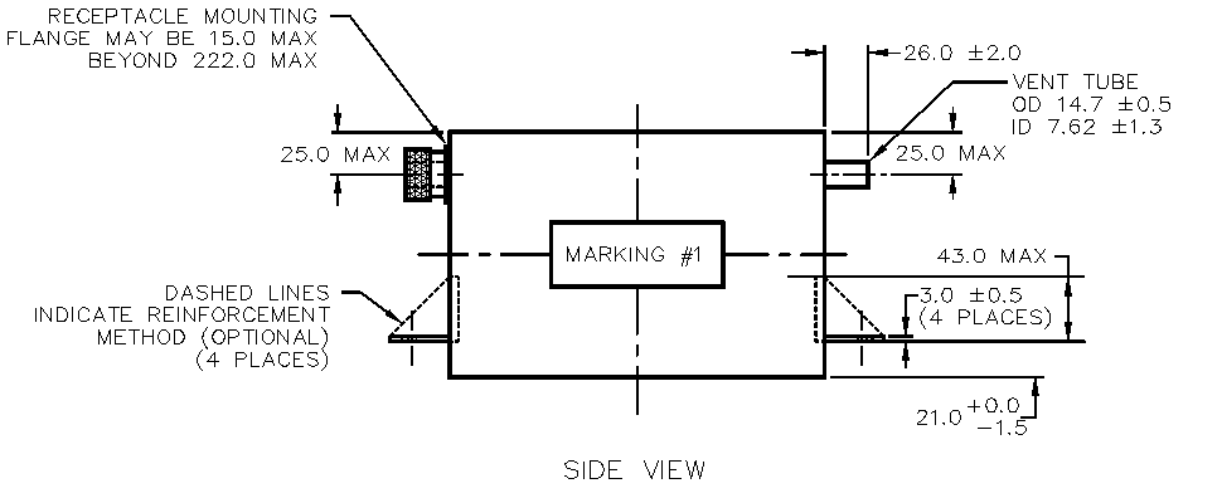


FIGURE 1. Dimensions and configuration for PIN M29595/1-1 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

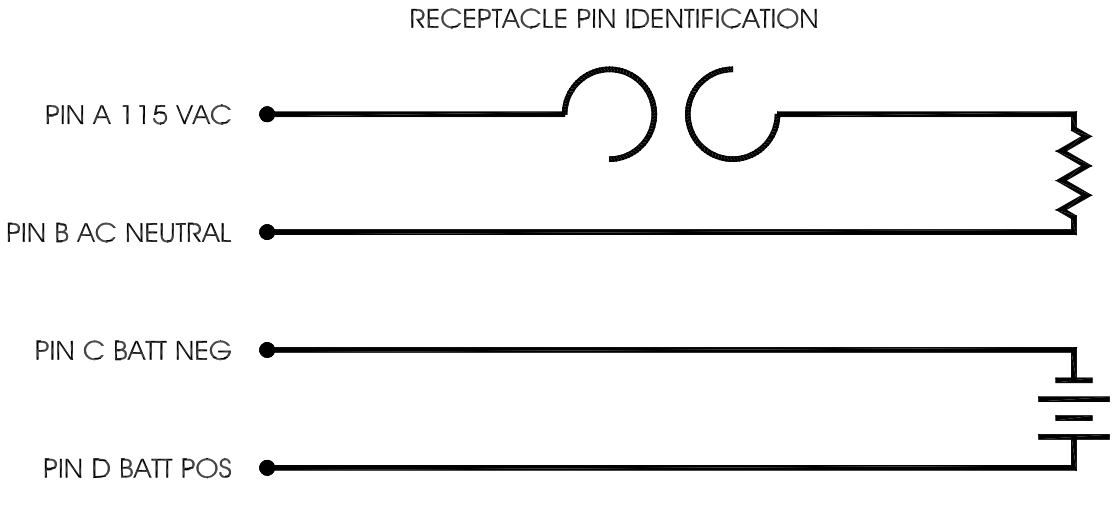


FIGURE 2. Electrical schematic for marking #3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

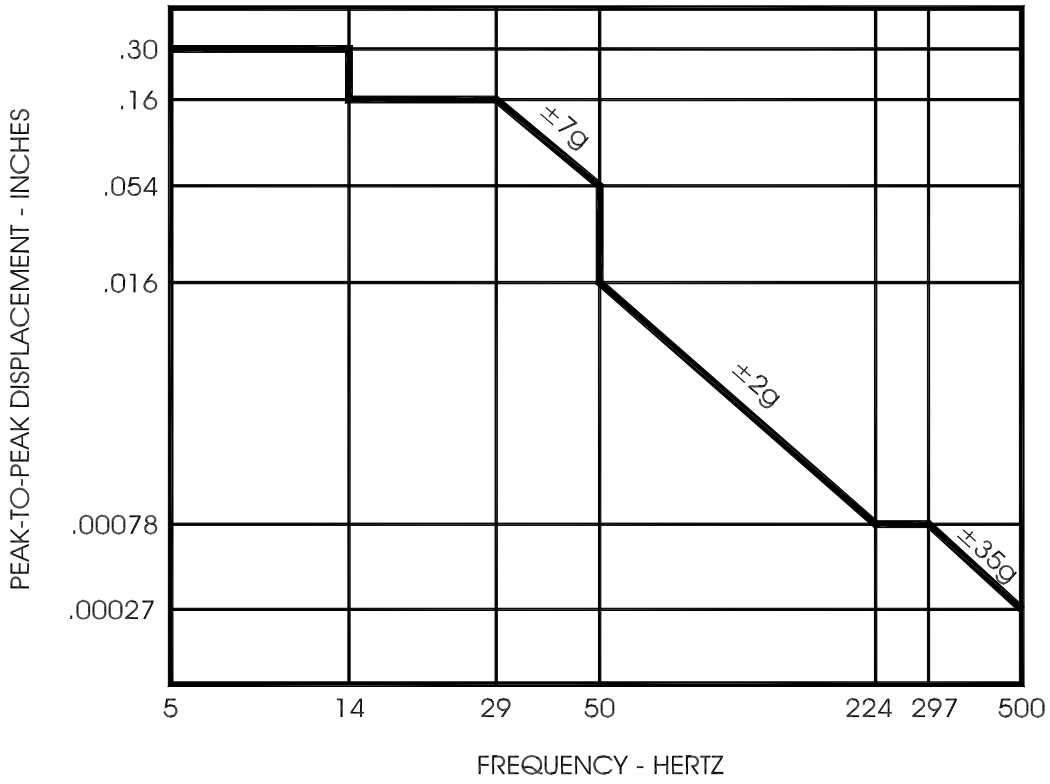


FIGURE 3. Sine wave vibration test levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

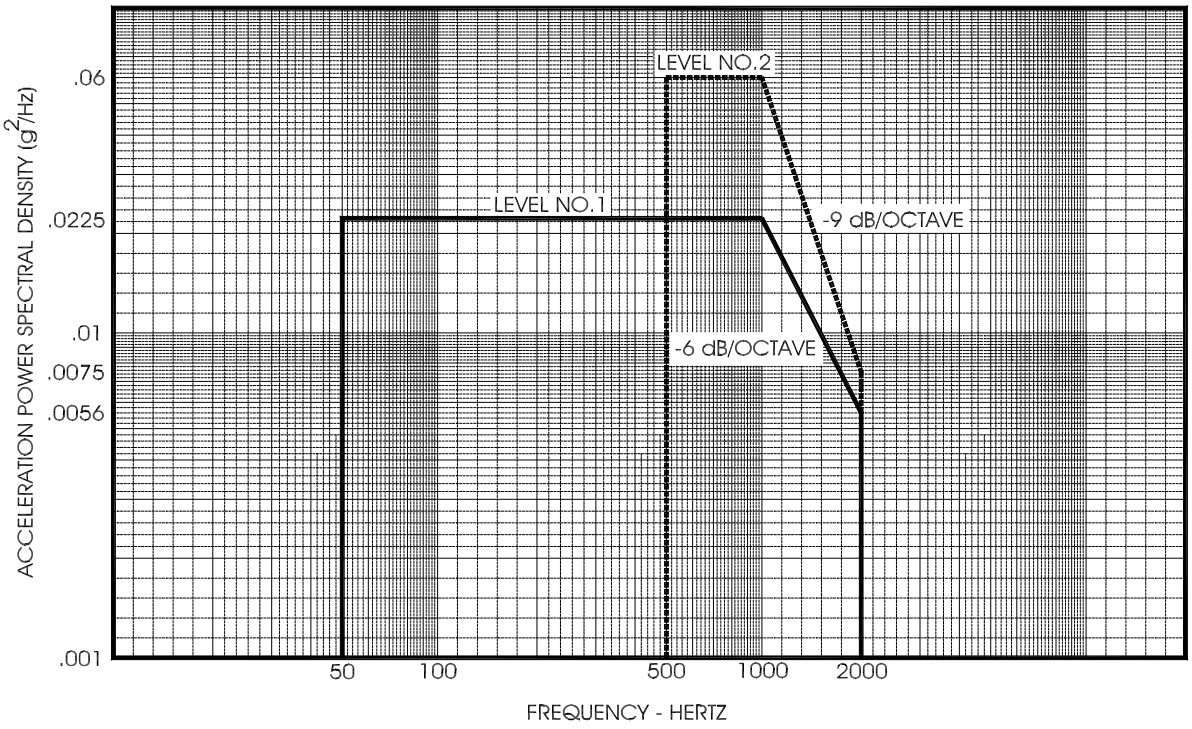


FIGURE 4. Random vibration test levels.

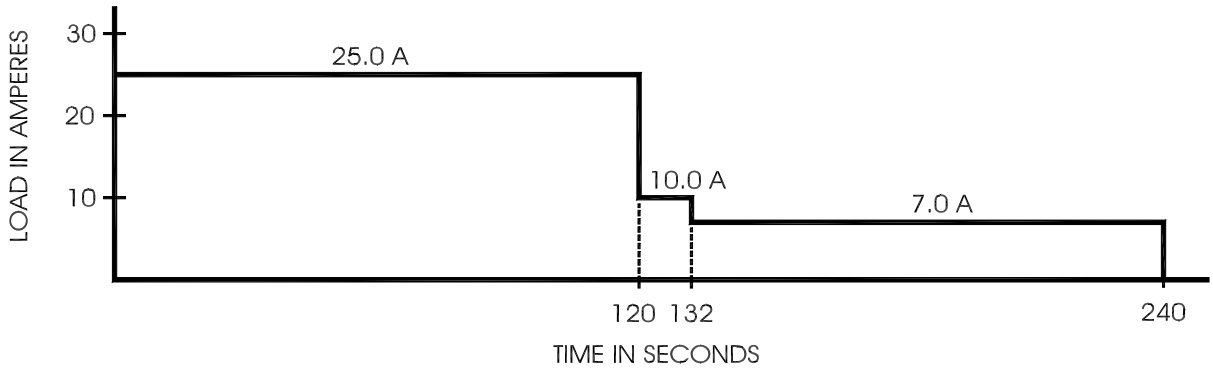
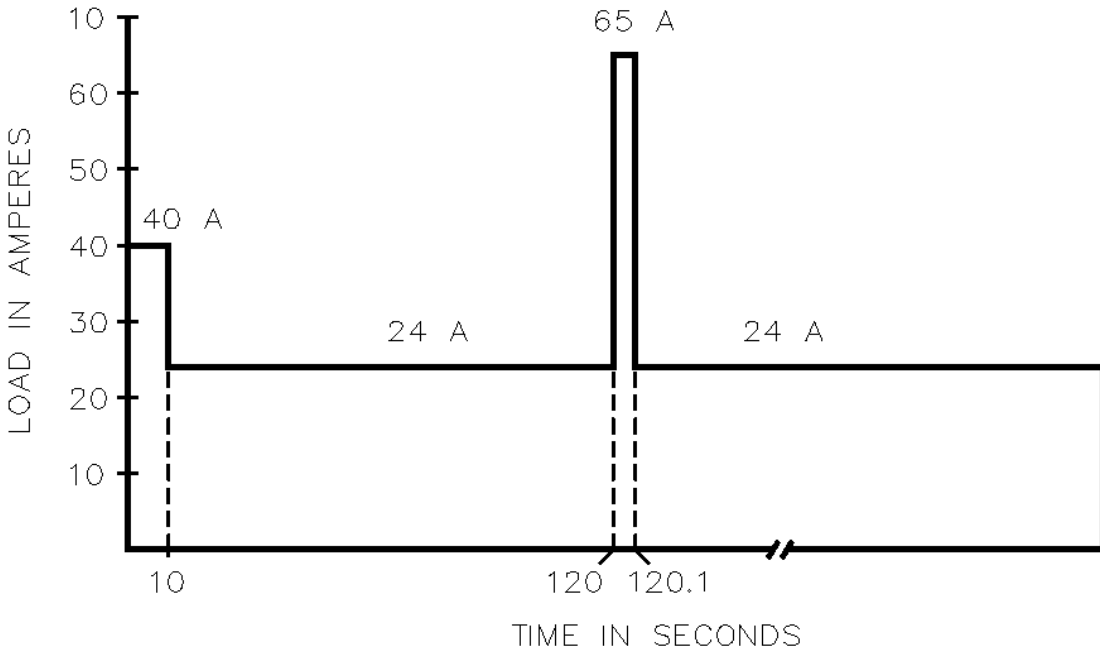


FIGURE 5. Aircraft start-up load requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.



Step	Sample number	Discharge temperature (°C)	Load current (amperes)	Discharge time (seconds)
	1, 2, 3	25	40 24 65 24	10 110 0.1 3600
2	1	-29	40 24 65 24	10 110 0.1 2700
3	2	-18	40 24 65 24	10 110 0.1 3000
4	3	0	40 24 65 24	10 110 0.1 3300
5	1	55	40 24 65 24	10 110 0.1 3600

FIGURE 6. Aircraft emergency load requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing activity:
Navy - AS

Agent:
Navy – SH

(Project 6140-2011-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:29Z
Check the source to verify that this is the current version before use.